THE VIEWER — visual changelog (complete, auto-generated)

129 releases · generated from docs/CHANGELOG.md — rerun _make_changelog_visual_full.py after any release · R3/R5

🟩 feature 🟫 fix 🟦 rule / structure

0.1.0 Architecture & diagrams (Milestone M1)
2026-06-01

Added

Master architecture (docs/ARCHITECTURE.md), decision log (docs/DECISIONS.md). Architecture & data-flow diagrams (system, ingestion queue, search, data model) as Mermaid sources

0.2.0 Indexing engine + OCR pipeline
2026-06-01

Added

engine/viewer_ingest.py: crawl / extract / OCR / status / search: text-first, resumable, Migration 0001_init.sql: versioned schema — documents, pages, FTS5 pages_fts (+ sync

0.3.0 Onboarding + 104th parts request
2026-06-01

Added

Migration 0002_sessions.sql: sessions, faults, request_items (additive; schema v2). Offline web app engine/viewer_app.py (/api/search, /api/request) + dark single-page UI

0.4.0 Full-corpus extraction + durability
2026-06-01

Added — Fixed

Resumable, time-boxed batch crawling (--max-files, --max-seconds); cheap skip (size+mtime Index corruption when a write was interrupted on the bridge filesystem: VIEWER_RELAXED mode

0.5.0 PyMuPDF text backend
2026-06-01

Changed

PDF text extraction switched to PyMuPDF (pymupdf, pip) with Poppler pdftotext fallback —

0.6.0 RapidOCR engine; full index run on PC
2026-06-01

Changed — Added

OCR engine switched to RapidOCR (rapidocr-onnxruntime, pip, no admin, bundled models). Full text-first crawl launched and completed on the Windows PC: 40,291 documents,

0.6.1 OCR detail + example
2026-06-01

Added

Detailed OCR diagram docs/diagrams/07-ocr-detailed; OCR before/after example PDF OCR build/feature map docs/diagrams/06-ocr-build-featureset.

0.7.0 Dark diagram standard (R3)
2026-06-01

Added

Standing rule R3: all diagrams use a professional dark theme and ship a PDF. Dark palette + cairosvg SVG→PDF export. Converted 06/07 to dark + PDF; consolidated

0.8.0 Search GUI: document viewer
2026-06-01

Added

GET /page?doc=&page=&dpi= renders any PDF page to PNG on demand (PyMuPDF); GET /api/doc metadata. UI document viewer: click a result → full-screen real manual page, prev/next + arrow/Esc keys,

0.8.1 OCR reliability fixes
2026-06-01

Fixed — Changed

OCR was failing 100% on the PC: the queue's first rows were leftover sandbox-path docs OCR parallelism switched from ProcessPoolExecutor to ThreadPoolExecutor (shared engine; PyMuPDF

0.9.0 Forked: GPU production + Lite portable
2026-06-01

Added — Notes

Engine flags --gpu (RapidOCR on CUDA via onnxruntime-gpu, automatic CPU fallback) and --dpi Advanced/GPU production build (master): engine/run_ocr_gpu.bat, docs/SETUP-GPU.md.

0.9.1 Changelog standard (R4)
2026-06-01

Added

Standing rule R4: a changelog entry accompanies every change. This docs/CHANGELOG.md, retroactively covering all prior builds.

0.9.2 Visual changelog standard (R5)
2026-06-01

Added

Standing rule R5: every changelog entry also gets a detailed graphical explanation and a docs/diagrams/CHANGELOG-VISUAL.svg + .pdf — a dark, retroactive visual changelog with a

0.9.3 Launcher pip self-upgrade
2026-06-01

Fixed — Added

Every setup/launch file now runs %PY% -m pip install --upgrade pip before installing packages. Affected: run_indexing.bat, run_ocr.bat, run_ocr_gpu.bat, run_app.bat, and the Lite

0.9.4 Part nomenclature carries to the sheet
2026-06-01

Fixed

The cart's Item Name started blank, so a part's nomenclature didn't reach the 104th sheet Verified end-to-end: derived names (e.g. "OVERHAUL GASKET SET") now appear in the ITEM NAME blocks

0.10.0 NSN search + Vehicle breakdown hub
2026-06-01

Added — Notes — Verified

Full-NSN search (part + vehicle): the server detects a full NSN in the query, exact-matches it Vehicle breakdown hub (GET /api/vehicle?key=): a vehicle NSN (or clicking a vehicle name)

0.11.0 Express modal + Schematics on every page
2026-06-01

Added — Notes — Verified

Modal express search / bypass: an always-on "What do you need?" field (part name / part # / NSN) Always-on "Schematics & install" panel in the document viewer (fallback chain, never empty):

0.12.0 OCR speedup: skip-the-junk + prioritized queue
2026-06-01

Added — Verified — Note

Skip-the-junk: ocr_one now checks each page for blankness (render @50dpi, ink ratio) and Prioritized OCR queue: new prioritize command (auto-run by ocrall) orders OCR — parts

0.13.0 Dynamic front-end: home, smart results, viewer zoom/thumbnails/highlight, responsive
2026-06-01

Added — Notes — Verified

Home / browse-by-vehicle: the search column now shows a home screen when the box is empty — a Smart results: results gain a filter bar with live counts ("N of M · K vehicles") and clickable

0.14.0 Search upgrades: "Last-4" NSN lookup + smarter key terms
2026-06-01

Added — Notes — Verified

"Last 4" lookup: typing exactly 4 digits (e.g. 2202) now finds every manual whose Synonym / alias expansion: an extensible engine/synonyms.json (32 starter groups) means one term

0.15.0 Onboarding modal aligned to the 104th sheet header
2026-06-01

Changed — Added — Notes

The onboarding modal now mirrors the 104th ECC Parts Request Sheet header 1:1 — same field Header fields are always visible. Previously "Simple view" hid UOC / Tech status / Motor sergeant,

0.16.0 Self-learning search + calmer modal/search
2026-06-01

Added — Changed — Notes — Verified

Learns from successful sheets. A new GET /api/popular-parts endpoint (every part that Rotating example. The search box and modal field now show a single, real example that rotates

0.17.0 Tech Status derived from the fault + part (PMCS-cited), mandatory at export
2026-06-01

Added — Notes — Verified

GET /api/techstatus suggests an equipment status from the fault and requested parts: Mandatory confirm gate at export. "Export Parts Request Sheet" now opens a gate that shows the

0.18.0 Switchable layouts behind a Settings panel (presentation only)
2026-06-01

Added — Notes — Changed — Verified

⚙ Settings panel (header) that consolidates the previously scattered toggles into one place: Named layout presets that bundle those settings: Simple/Junior, Advanced/SME,

0.19.0 Structured parts index (Phase 1) + coverage meter + quick wins
2026-06-01

Added — Notes — Verified

RPSTL parts extractor (viewer_ingest.py parts): scans parts catalogs ("Usable On Code" pages) GET /api/part?nsn= — cited catalog references for an NSN (which figures/pages/vehicles it

0.20.0 Online → offline reference enrichment (cited, official-only)
2026-06-01

Added — Notes — Verified

viewer_ingest.py enrich — a one-time online enrichment that the engine then uses offline: GET /api/reference?nsn=&size= and a cart line: "▯ External reference (cited, offline): …"

0.21.0 Append-only NSN enrichment (rule R6) + dataset-growth scope
2026-06-01

Added — Notes — Verified

Standing rule R6 — append-only data: you may always ADD to the search engine, but never take away, Append-only NSN reference log (migration 0006, ref_nsn_log): every GSA enrichment pass now

0.21.1 One-time enrichment reads the real GSA file format (XLSX)
2026-06-01

Added — Notes — Verified

enrich --gsa now reads the official GSA extract directly as .xlsx (streaming via openpyxl) as Confirms the intended posture: the running engine never goes online; enrich is a one-shot,

0.22.0 PUB LOG reference ingest (proof of concept)
2026-06-01

Added — Why this matters — Notes — Verified

enrich --publog <csv|xlsx> ingests a PUB LOG (DLA) export — the authoritative, public-releasable Migration 0007: ref_nsn / ref_nsn_log gain part_no, cagec, characteristics, aac,

0.22.1 One-click enrichment launcher + PUB LOG export quickstart
2026-06-01

Added — Note

engine/run_enrich.bat — drag a PUB LOG export (CSV/XLSX) onto it (or pass a path) to run the one-time docs/PUBLOG-EXPORT-QUICKSTART.md — exact steps to download PUB LOG (free, no CAC), export the right

0.23.0 PUB LOG via the FLIS Reading Room (direct CSVs, no Windows app)
2026-06-01

Added — Notes — Verified

enrich --publog-dir <folder> ingests the DLA FLIS Data Electronic Reading Room files directly: The ingester reads every CSV/XLSX in the folder, keeps only in-index NSNs, composes the NSN from

0.24.0 PUB LOG enrichment RUN on the live index (NIIN-keyed FLIS ingest)
2026-06-02

Done — Added — Notes — Verified

Ingested the user's downloaded DLA FLIS Reading Room catalog (~16 GB of table extracts) into the Real examples now offline & cited: 6115-01-036-6374 → GENERATOR SET, DIESEL ENGINE · P/N MEP007B ·

0.25.0 Overnight: full-catalog parts + FLIS enrichment, supersession/vintage, 2D→3D, rollback
2026-06-02

Done — Added — Search speed — OCR

Structured parts on the full index: extracted 227,908 records / 45,068 distinct part NSNs FLIS enrichment of the whole catalog: 41,701 NSNs filled from the DLA FLIS Reading Room

0.26.0 3D viewer upgrade: family shapes, material/colour, expanded dimensions
2026-06-02

Changed — Verified

The representative 3D viewer now renders family shapes (cylinder · hex · disc · box) chosen from the Material / colour / finish appearance, grounded in FLIS: the solid is tinted to the stated colour

0.27.0 Schematic legibility viewer: Clean + contrast, 3D tilt, hover loupe
2026-06-02

Added — Invariants — Verified — Next

Clean toggle + contrast slider in the page viewer. Server-side, the real page is re-rendered hi-DPI 3D tilt of the flat schematic — a CSS perspective rotateY on the page image (−45°…+45°), to read a

0.28.0 Hi-fi loupe + dataset correlations + pillar & mutation tests
2026-06-02

Added — Invariants — Verified

High-fidelity loupe. /page now accepts a clip=x0,y0,x1,y1 sub-rectangle and renders just Correlations sidecar (index/correlations.db). A new, fully additive read-only DB derived from

0.29.0 Data protection: integrity safeguard, snapshot vault, recovery + hardened tests
2026-06-02

Root cause — Added — Verified

Reproduced and characterised: the truncation was a sandbox read-cache artifact at the host→guest engine/safeguard.py — integrity + recovery ("the treasure vault"). Atomic writes

0.30.0 Schematic orientation: dual-axis tilt, mirror with readable labels, on-demand HD
2026-06-02

Added — Invariants — Verified

Dual-axis tilt. The viewer now tilts the page on both axes — tilt Y (left/right) and tilt X Mirror (↔) with readable labels. A horizontal flip to orient from the opposite side of the

0.31.0 System Status page, automatic snapshots, OCR finishing, recall/curation features
2026-06-02

Added — Honest gaps — Verified

System Status page (/status, /api/status). One-glance health: document/page/parts/NSN counts, Automatic snapshots. register_snapshot_task.bat registers a daily Windows Scheduled Task

0.32.0 NIIN-review confirm/reject workflow + OCR run guide
2026-06-02

Added — Verified

NIIN-drift review workflow (actionable). The 884 drift groups are now a working queue on the OCR run guide (docs/OCR-RUN-GUIDE.md): preflight checklist, step-by-step run_ocr_gpu.bat run,

0.33.0 Interchangeable-NSN alias map in search + GPU readiness + queue prioritized
2026-06-02

Added — Done — Verified

Confirmed-interchangeable alias map. When you mark a NIIN-drift group interchangeable on the engine/gpu_check.py — one-command GPU readiness verdict (driver, onnxruntime providers, RapidOCR)

0.34.0 Tight, seamless loupe + cohesive accessibility controls
2026-06-02

Changed — Invariants — Verified

Loupe responsiveness reworked. The magnifier now tracks the cursor with zero latency: an Cohesive control feel. All viewer toggles (Clean, contrast, tilt Y/X, Mirror, HD, Loupe) share one

0.35.0 Page zoom: slider + scroll-to-cursor
2026-06-02

Added — Invariants — Verified

Zoom slider (100%–400%) in the viewer toolbar, alongside the tilt X/Y sliders — slide the Scroll-to-zoom toward the cursor when the loupe is off: hover a spot and scroll to zoom in/out

0.36.0 Hardware probe + autonomous adaptive GPU OCR (Win11-first, back to Win7)
2026-06-02

Added — Honest scope — Verified

Capability probe (engine/sysprobe.py): detects OS+build (Windows 7→11), Python, CPU cores, RAM, Autonomous adaptive OCR runner (engine/run_ocr_auto.bat): probes → installs the right stack

0.37.0 COMPLETE backward compatibility: Windows 11 → Vista
2026-06-02

Changed — Added — Honest scope — Verified

Reframed from "incomplete/best-effort" to complete feature compatibility from Windows 11 down to Poppler render fallback in viewer_app.py: when PyMuPDF can't be installed (legacy OS), the

0.38.0 Dual-track changelog scaffolding (legacy track, rule R7)
2026-06-02

Added — Notes — Verified

Rule R7 — dual-track changelog. Legacy (Retroactive Post-Support) builds get their own changelog Concept mockup: docs/diagrams/53-dualtrack-changelog-mockup.pdf (the approved format).

0.39.0 3D Library + Schematics Library + Reset on every moveable view
2026-06-02

Added — Invariants — Verified

Reset buttons (the missing control). Every moveable/interactive view now has a clearly-labelled ▯ 3D Library (/3d, /api/thread). A dedicated, searchable, paginated gallery of all 20,869

0.40.0 Make it pop: real-time WebGL 3D + dynamic schematics (current-specs)
2026-06-02

Added — Invariants — Verified — Next

engine/ui/gl3d.js — a dependency-free WebGL 3D renderer (no Three.js, no CDN → fully offline). Dynamic schematics viewer: buttery pan (drag) + cursor-centered wheel zoom (transform-

0.41.0 Schematics: explore from every angle + related-sheets rail
2026-06-02

Added — Invariants — Verified

Tilt to any angle + mirror in the dedicated /schematics viewer: tilt X and tilt Y sliders Related-sheets rail: a filmstrip of the same vehicle's other schematic / wiring sheets

0.42.0 Circuit Lab: overlay editor + real-time circuit simulator
2026-06-02

Added — Invariants — Verified

engine/ui/circuitsim.js — a dependency-free analog circuit engine (Modified Nodal Analysis). engine/ui/circuitlab.html — the Circuit Lab (/circuitlab): a schematic-overlay editor + live

0.43.0 Look-Alike Parts recognizer
2026-06-02

Added — Invariants — Verified

part_differences() + /api/partdiff + the /partdiff page — a recognizer for parts that Four colour-coded verdicts per variant: reference (what you searched), different variant

0.44.0 Circuit Lab, deepened: active devices + save/load/export + part links
2026-06-02

Added — Invariants — Verified

Four new active devices in the MNA engine (circuitsim.js), each unit-tested: AC source Generalised N-pin editor model: components carry a pin-offset table instead of fixed 2 leads, so

0.45.0 Retroactive Post-Support (RPS): comparable speed on older/slower PCs
2026-06-02

Added — Invariants — Verified

engine/rps.py — the RPS runtime layer. Picks a mode (modern / lite / legacy) from the Page-render disk cache (index/pagecache/): a full-page PNG is rendered once then served from

0.46.0 Faster transport (gzip + keep-alive) + a Performance toggle in Settings
2026-06-02

Added — Invariants — Verified

gzip + HTTP/1.1 keep-alive in the server (_send): the TCP connection is reused across requests. Performance (RPS) toggle in ⚙ Settings: Auto / Modern / Lite / Legacy, saved per-browser

0.47.0 How to do it: the procedure view
2026-06-02

Added — Invariants — Verified

procedure_for() + /api/procedure + the /procedure page — surfaces the step-by-step TM shows kind-tagged cards (Removal/Installation/... colour-coded), tools as chips you can stage,

0.48.0 Solve it: the workflow hub
2026-06-02

Added — Invariants — Verified

The /solve workflow hub — one screen that walks from a symptom to a fix: enter a problem Additive & grounded (R1/R6): pure client-side orchestration of endpoints that are already tested —

0.49.0 Type-ahead predictive search (offline)
2026-06-02

Added — Invariants — Verified

suggest() + /api/suggest + a search-box dropdown — google-style suggestions as you type, fully The dropdown is debounced (120 ms), keyboard-navigable (↑/↓ to move, Enter to pick, Esc to close),

0.50.0 Add documents: index new TMs without the command line
2026-06-02

Added — Invariants — Verified

The /ingest page + ingest_preview() / ingest_start() / ingest_status() — point THE VIEWER at Additive & safe (R1/R6): the crawl only adds documents + extracted text — it never deletes or

0.51.0 Printable job packet (take-to-the-bay)
2026-06-02

Added — Invariants — Verified

The /packet page — turns a part + its procedure into a print-optimised job sheet a mechanic engine/tests/test_features.py — a 21-test regression suite (procedure parser, procedure_for,

0.52.0 Unified part dossier
2026-06-02

Added — Invariants — Verified

The /dossier page — one page per NSN that aggregates everything we hold about a part: Read-only aggregation (R1/R6) of endpoints that already exist; the manual stays the source of truth.

0.53.0 Find in manual (in-document Ctrl+F)
2026-06-02

Added — Invariants — Verified

find_in_doc() + /api/findindoc + a viewer find box — the in-document Ctrl+F the mission Read-only; resets when a new document opens. Additive to the viewer only (R1/R6).

0.54.0 Ops / health dashboard
2026-06-02

Added — Invariants — Verified

/ops + ops_summary() + file_audit() — one glance at engine health: runtime RPS mode, Read-only & cheap (R1/R6): ops_summary() uses only light queries (no full OCR scan — that stays on

0.55.0 UX consolidation: visual steps, torque, command palette, help
2026-06-02

Added — Invariants — Verified

Visual steps (/stepflow) — the parsed procedure rendered as a big follow-along flow: numbered Torque specs — integrated (torque_specs() + /api/torque) — parses stated torque values

0.56.0 Speed & efficiency pass (Tier 1 + Tier 2)
2026-06-02

Added / Changed — Invariants — Verified

Open-PDF LRU cache — render_page_png re-opened and re-parsed the whole PDF on every page render. Thread-local DB connections — db() opened a new SQLite connection and re-applied PRAGMAs on every

0.57.0 Speed pass, round 2 (each change measured)
2026-06-02

Added / Changed — Invariants — Verified

Compact JSON — _send serializes with separators=(",",":") (no inter-token spaces). Measured: Pre-render page ETag — the /page route computes a cheap param-based ETag and checks

0.58.0 Gap-closing optimization pass
2026-06-02

Added / Changed — Invariants — Verified

OCR build — identical-page dedup + adaptive DPI (viewer_ingest.py) — a density probe skips blank Legacy memory + cold-start warmup (rps.py, viewer_app.py) — the open-PDF LRU is now sized per RPS

0.59.0 OCR-driven Smart Collections (Batch 1 of 3)
2026-06-02

Added — Invariants — Verified

Smart Collections (/collections, viewer_app.py, ui/collections.html) — a collection is a saved APIs — GET /api/collections lists every collection with a bounded live count (COUNT over a

0.60.0 OCR-driven page callouts (Batch 2 of 3)
2026-06-02

Added — Invariants — Verified

Page callouts (page_callouts() + GET /api/callouts, viewer_app.py) — extracts the high-precision Positioned hotspots + chip bar (index.html) — a new ▯ Callouts toggle in the viewer. Where the page

0.61.0 Smart Collections: four add-ons
2026-06-02

Added — Invariants — Verified

Scope a collection to a vehicle and/or manual-type — a saved collection can be limited to one vehicle "New since last visit" badge — opening a collection records its current size in a collection_seen

0.62.0 3D library wired to the manuals (Batch 3 of 3)
2026-06-02

Added — Invariants — Verified

3D part → manual references (thread_refs() + GET /api/threed_refs, viewer_app.py) — opening a Collections membership — for each Smart Collection, a cheap EXISTS asks whether any page matches both

0.63.0 RPS-safe stability suite (health layer)
2026-06-02

Added — Invariants — Verified

Preflight health gate (preflight.py) — before the server or OCR start, checks python / free disk / Disk-space guard (disk_ok()) in preflight.py — watches free space on the index drive (default 1 GB

0.64.0 Foundation batch (part 1): the test + RPS gate
2026-06-03

Added — Fixed — Verified — Next

RPS lint (tests/rps_lint.py) — the "does legacy still work?" gate. Scans every ui/.html + /healthz — new GET endpoint returning the preflight checks as JSON (python/disk/DB integrity/schema/GPU);

0.65.0 Foundation batch (part 2): shared code + route smoke test
2026-06-03

Added — Verified — Deferred

patterns.py — one home for the NSN / FIG / labelled part-number regexes + norm_nsn / digits / docs/diagrams/_common.py — shared diagram helpers (palette + box/tiwrap/panel/svg_open/render)

0.66.0 OCR scan COMPLETE — finalize the full corpus
2026-06-03

Context — Added — Why host-side — Verified

The pages_fts trigger means search, find-in-manual, Smart Collections (auto-fill + "new" badges), FINALIZE-OCR.bat — one host-side button that locks in the complete scan, in order: (1) re-extract the

0.67.0 PUB LOG cross-reference, repository mode, and a keyword/tag layer
2026-06-03

Added — Added — Added — Invariants

Extended enrich_flis to tap data it ignored, marking every NSN in the index against the DLA FLIS ENRICH-PUBLOG.BAT — one host-side button that runs the full-folder cross-reference

0.68.0 3D viewer: detailed parametric part models (no more blocks)
2026-06-03

Added — Invariants — Verified

ui/partgeo.js — a dependency-free geometry library that builds detailed part meshes: a bolt with Wired into threed.html: the gallery thumbnails stay light (fast grid), and opening a part upgrades it to

0.69.0 Inline part tagging (background pencil), not a foreground feature
2026-06-03

Added — Changed — Verified

ui/tagger.js — a dependency-free, ES5-safe inline tagger. A quiet pencil icon marks a part as Pencil on every part while browsing — on search-result cards that carry an NSN, and on every 3D

0.70.0 Engine modularization (begun) + truncation cleared
2026-06-03

Changed — Verified

Extracted Smart Collections into collections_feature.py — the largest cohesive block now lives in its The engine was already multi-module (rps, preflight, patterns, safeguard, sysprobe,

0.70.1 Fix: app wouldn't start (circular import) + schematic path extractor
2026-06-03

Fixed — Added

The 0.70.0 modularization made collections_feature.py do import viewer_app as core. That's fine when the schem_overlay.py — schem_paths(pdf, page) reads a vector schematic's drawing ops + word boxes via

0.71.0 Schematic Highlighter (Phase 1) + end-to-end demo/test suite
2026-06-03

Added — Added — Verified

Clickable vector overlay — on a vector schematic, the new ▯ Highlight toggle replaces the static Raster fallback — scanned sheets (no vector geometry) show the existing callout chips (NSN/part/figure

0.72.0 Reconstituted Fix/procedure pages (all 4 + both exports)
2026-06-03

Added — Verified

procedure_feature.py — a deepened parser + correlation engine (new module, DI-injected core, no Rebuilt procedure.html (ES5-safe) delivering all four requested pieces:

0.72.1 Answer: most-common part nomenclature (host-side ranking)
2026-06-03

Added — Why host-side — Verified — Compatibility

engine/top_nomenclature.py upgraded — now emits three views and saves them to index/MOST-COMMON-PART.txt: ANSWER-MOST-COMMON-PART.bat — one double-click: runs the ranking on the live index, prints the answer,

0.72.2 Fix: /api/threed_refs 500 (orphaned `_collections_defs` after modularization)
2026-06-03

Fixed — Why the other tests w... — Verified — Compatibility

/api/threed_refs returned HTTP 500 whenever a valid NSN was passed. When the 3D→manual hookup tried to Fix: added _collections_defs to the from collections_feature import (...) line in viewer_app.py.

0.72.3 Audit pass: preflight hang fix + connection-leak fixes + mutation harness
2026-06-03

Fixed — Recovered — Added — Measured

Preflight hang on startup. preflight.py's index check ran a full PRAGMA quick_check, which reads Connection leak in procedure_full (procedure_feature.py). Opened core.db() and never closed it —

0.73.0 Two sides of the house: Operator (10) vs Mechanic (20)
2026-06-03

Added — Design notes — Verified — Compatibility

Authoritative tm_side() classifier (patterns.py). Divides every document into the operator /api/by_side?side=operator|mechanic — lists the documents on a side, with live counts

0.74.0 Tighter sorting: side-map cache + manual overrides + cover/MAC corroboration
2026-06-03

Added — Added — Planned — Verified

Side-map cache (speed). Every document is classified once and cached, keyed on a cheap signature Manual overrides (accuracy). sides_override.json sidecar (append-only, keeps a log) lets you pin any

0.75.0 Phase 2: chapter-level routing inside combined manuals
2026-06-03

Added — Added — Verified — Compatibility

Chapter→side ranges. For combined manuals only, scans the OCR'd page text for chapter/section headings Lazy + cached + persisted. Ranges are computed on first open and cached per doc (keyed on page count),

0.75.1 Chapter-split review UI
2026-06-03

Added — Verified — Compatibility

GET /api/chapters_review — lists every combined manual with its detected sections, per-side landing "Review chapter splits" panel in the side chooser — lists combined manuals with their split status;

0.76.0 Real part imagery: cited figure crops in 3D previews + "Manual illustration"
2026-06-03

Why — Added — Added — Added

figure_for(nsn) → the best citing figure (doc/page/fig) from the parts index. Figure-region crop (extract / get_crop): PyMuPDF renders just the figure region of the cited page —

0.77.0 Scanned-page caption tightening + experimental local image→3D scaffold
2026-06-03

Tightened — Added — Why experimental / ho... — Compatibility

OCR word-box caption anchor — on-demand OCR (pytesseract, optional) finds the "FIGURE n" caption and Row ink-density fallback — if OCR isn't available, a numpy row-darkness profile finds where the dense

0.78.0 Part-number ↔ figure correlation (RPSTL rows + breakdown image)
2026-06-03

Added — Added — Added — Tiers delivered

parse_line/parse_page turn the parts-list table into structured rows — item · SMR · NSN · CAGEC · lookup(pn, cagec) resolves a part number to its row + the breakdown image URL (/figcrop) + a

0.78.1 Part# review panel + breakdown image in the dossier
2026-06-03

▯ Part# review — Dossier — Breakdown image

▯ Part# review (header) — an overlay listing low-confidence RPSTL rows with "Open" (jump to the page) Dossier now shows a Breakdown image card (the part's cited figure crop) alongside catalog refs,

0.79.0 Cross-reference & verification engine (X1–X5)
2026-06-03

Added — Added — Added — UI

part_record(pn|nsn) unifies everything we hold offline: the RPSTL row + FLIS official name /api/part_record, /api/xref_coverage.

0.79.1 xref coverage in Ops + cross-reference in procedure & 3-D
2026-06-03

Ops dashboard — Part-number cross-ref... — in review — 3-D modal

Ops dashboard gains a Part-number cross-reference card: parsed rows · with NSN (%) · FLIS-validated 3-D modal side panel gains a CROSS-REFERENCE block for the opened part — FLIS name (OCR conflict

0.80.0 Map the scan's colour + material onto the 3-D models
2026-06-03

Added — Changed — Honest scope — Verified

material_for(characteristics, name) parses the scan's COLOR, MATERIAL and SURFACE /api/part_material?nsn=|chars=|name= — reads FLIS characteristics (ref_nsn) when only an NSN is given.

0.81.0 Live parametric CAD panel (procedural geometry, editable)
2026-06-03

Context — Added — Changed — Verified

PARAMETRIC CAD panel in the modal: a Family selector + editable Length / Width / Height / Values are cited from FLIS but editable — the mesh is generated, not fixed.

0.81.1 Shape + colour + material on EVERY 3-D representation
2026-06-03

Changed — Changed — Verified — Compatibility

The client appearance() was upgraded from a 4-colour stub to the full colour + material + finish So every collection-card thumbnail and the SVG fallback now show the part in its actual colour + finish

0.81.2 One-click installer for the 3-D / imagery prerequisites
2026-06-03

`INSTALL-3D-DEPS.bat` — PyMuPDF, Pillow, nump... — `engine/verify_3d_dep... — Tesseract

INSTALL-3D-DEPS.bat — installs the imagery pipeline's Python deps (PyMuPDF, Pillow, numpy, engine/verify_3d_deps.py — reports each prerequisite as OK/missing and what it enables; detects the

0.81.3 Silence harmless MuPDF colour-profile noise
2026-06-03

still renders the page

Some scanned PDFs carry a broken embedded ICC colour profile; MuPDF logs cmsOpenProfileFromMem failed, No PDF is modified (corpus stays read-only); the messages were noise, not errors.

0.81.4 3-D library shows the real cited figure + stop stale-JS caching
2026-06-03

Changed — Operational note

/api/threed now resolves each part's cited figure server-side and includes image_url + fig_no, so partgeo.js and gl3d.js are now served Cache-Control: no-cache (was max-age=3600). The hour-long

0.81.5 Diagnosed the "all squares" report: empty FLIS names + browser cache
2026-06-03

What the diagnostic f... — Fixed — Operational

Cited figure FOUND 15/15; crop PNG on disk 15/15 — the figure pipeline is correct and complete; the shape=box 12/15 — root cause: ref_nsn.item_name is empty for many parts, so the shape classifier

0.82.0 Figures-first 3-D library: lead with the parts that have a real manual image
2026-06-03

Changed — UI — Compatibility — Operational

threed_list() now defaults to figures-only. The 3-D library is sourced from the parts table where a /api/threed honors ?all=1 to fall back to the full FLIS-dimension set (representative parametric shapes,

0.83.0 Interactive demo / onboarding tour (both sides), one double-click
2026-06-04

Added — Why it's robust — Compatibility

engine/ui/demo.html — a single, self-contained guided demo (pure ES5, dark theme R3, zero external DEMO.bat — double-click to open the demo in the default browser. No server, no install.

0.83.1 Demo becomes the onboarding intro; "Finish" hands off to the real side chooser
2026-06-04

Changed — demo.html — Compatibility

First-run flow (index.html): a new first-run user now sees the guided tour first (the demo loaded in a "Finish" / "Skip" hand-off: when the tour ends, the demo (in the iframe) posts a message to the app, which

0.83.2 Skip buttons: skip the demo, and skip straight to browsing
2026-06-04

Added — Compatibility

Skip the demo — a "Skip the demo →" button on the tour's opening gate (so you can bail before picking a Skip to browsing — a "Skip to browsing →" button placed next to "▶ Watch the guided tour" in the side

0.83.3 End-to-end verification harness + cache the heavy coverage stat
2026-06-04

Added — Fixed

RUN-E2E.bat + engine/diag_e2e.py — a full end-to-end smoke test. It starts a clean server on port 8766 Result of the first run: 48 / 49 checks passed. All 17 shell pages, all 8 JS assets, search (both sides),

0.84.0 Parametric shape pass: box-rate 24.7% → 9.3% on the real corpus
2026-06-04

Added — Fixed — Measured — Compatibility

CLAMP was misclassified as shaft because the pattern LAMP had no word boundary and matched "cLAMP". Routed the real top offenders to shapes: covers/doors/panels/guards → cover; markers/decals/labels/armor/

0.84.1 NSN/FSC shape fallback for nameless parts: box-rate 9.3% → 1.5%
2026-06-04

Added — Measured — Verified — Compatibility

familyFromNSN() + classify() in partgeo.js. When a part's name is missing, generic ("NAME", "ITEM", Name-only: 9.3% box. + NSN/FSC fallback: 1.5% box → 98.5% recognizable.

0.84.2 Detailed geometry in BOTH views + full 3-D wiring audit
2026-06-04

Changed — Verified — Compatibility

The card thumbnail (the drawn "artist approximation") now builds the same detailed PartGeo mesh Name classifier → 22 distinct families — all 22 have a builder.

0.85.0 Streamlined image search: number → results → figure/3-D/pages, + reusable loupe
2026-06-04

Added — Added — Routes — Compatibility

The ▯ Callouts markers on a TM page now open a right-side drawer (page stays visible) instead of jumping Page highlighting: clicking a number highlights it on the current page; clicking one of the "pages that hold

0.86.0 CAD images for the whole representative 3-D library (~20,869 parts)
2026-06-04

Added — Added — Wired — Verified

A pure-Python + Pillow renderer (no GPU, no heavy deps). For any part it classifies the shape (name + /cadimg?nsn= — renders on first request and caches (the figcrop pattern); subsequent requests are instant.

0.87.0 CAD roadmap items: real STL/OBJ export + fidelity pass + downloads
2026-06-04

Added — Improved — Operational — Verified

/cadstl?nsn= and /cadobj?nsn= — download the representative part as an actual STL (triangulated, ASCII) Download buttons: the 3-D modal now offers CAD image (PNG) · STL · OBJ; the part-assets drawer adds

0.88.0 CAD engine: wrap models in material textures + parse FLIS colours
2026-06-04

Added — Added — Changed — Verified

Each CAD render is now wrapped in a per-material surface texture (screen-space, masked to the part material_props() now parses the FLIS colour (OLIVE DRAB, CARC GREEN, FOREST GREEN, DESERT SAND/TAN, FIELD

0.88.1 CAD render `style` switch + comparison/contact-sheet tooling
2026-06-04

+ right-side-up + spe... — `make_contact.py` / `... — `make_compare.py` / `... — v1-vs-v2 comparison s...

cad_render.render(..., style="v1"|"v2"|"v3") — render any historical look on demand: v1 original (head-down, make_contact.py / RENDER-CONTACT.bat — a 10-part textured contact sheet (docs/cad_contact_sheet.png).

0.89.0 CAD detail level scales with the program build (RPS tier)
2026-06-04

Changed — Why — Verified

The CAD detail level now follows the RPS tier: modern → v3 (FLIS colour + material textures + specular), /cadimg resolves the style automatically from the server's RPS_MODE (cad_render.TIER_STYLE), with an

0.89.1 Failure-proof CAD renders + render all three tiers
2026-06-04

Hardened — Added — Compatibility

cad_render.render() now always yields an image for a known part: the texture step is wrapped (cosmetic, RUN-CAD-TIERS.bat — renders all three detail tiers in sequence: fills modern v3, then v2 (lite),

0.90.0 CAD-first 3-D library: the CAD image is now the face of every part
2026-06-04

Changed — Net effect — Verified

Every grid card now leads with the rendered CAD image (/cadimg), not the parametric SVG. The SVG stays only The part modal opens on the CAD image by default. Tabs reordered to ▯ CAD image · ▯ Interactive 3-D ·

0.91.0 "Living Schematic" (PoC): inferred netlist + animated flow overlay
2026-06-04

Added — Added — Added — Verified

Infers a connectivity graph (netlist) from a schematic page's vectors. Reuses schem_overlay.schem_paths Serves the inferred graph, cached to index/schemcache/<doc>_<page>.json. &fresh=1 bypasses the cache.

0.92.0 Interactive CAD: the CAD image now rotates, zooms, and scales like the 3-D model
2026-06-04

Added — Added — Added — Verified

render() gained yaw (spin about the vertical axis), pitch (camera tilt override), and title (drop the title render_spin() / ensure_spin(): render the same CAD pipeline at N viewpoints around a full 360° and pack

0.92.1 Rotate CAD now spins on real WebGL — exactly as smooth as the 3-D model
2026-06-04

Changed

The Rotate CAD view now renders through the same GL3D WebGL viewer and the same orbit/zoom controls as The sprite-sheet turntable (/cadspin + cadview.js) is kept as the legacy / no-WebGL fallback so the

0.93.0 Higher-quality CAD across all tiers + CAD vs 3-D now visually distinct
2026-06-04

Changed — Changed — Verified

Supersample 2× → 3× — noticeably cleaner anti-aliased edges on every render. Finer tessellation of the curved primitives (cylinder 28→48, tube 32→52, torus 30×16→48×26, sphere 16×12→

0.93.1 Max-quality CAD pass on every tier (CAD_VERSION 4 → 5)
2026-06-04

Changed — Verified

Supersample 3× → 4× (render large, LANCZOS-downsample) — the crispest anti-aliasing the renderer can do. Key + fill + ambient lighting — a soft fill light now lifts the shadow side of every part, so modeling reads

0.93.2 FLIS colour on EVERY CAD tier (CAD_VERSION 5 → 6)
2026-06-04

v1, v2 and v3 — Tier ladder still int... — coloured — coloured

Colour (FLIS stated colour, else a material-based tint) now renders on v1, v2 and v3 — previously it was Tier ladder still intact: v1 flat-matte coloured, v2 +specular/metallic coloured, v3 +procedural

0.93.3 Material TEXTURE on EVERY CAD tier too (CAD_VERSION 6 → 7)
2026-06-04

surface texture — v1, v2 and v3 — coloured and textured — Tier ladder kept

The per-material procedural surface texture (brushed metal / brass, rubber & cast grain, wood grain, plastic, render() gained a texturize flag (default = textured; False reproduces the old untextured look for the

0.94.0 Wire your own local 3-D models into the viewer (replace the placeholder)
2026-06-04

Added — Added — Changed — Verified

Looks for a user-provided model at index/models3d/<NSN>.obj or .stl (sidecar, R1/R6) and parses it to Status (exists/fmt/mesh_url/filename) and the parsed mesh JSON. Mirrors the image3d route style; 404 JSON

0.95.0 Hardware-aware boost: parallel CAD batch + CAD textures grafted onto the 3-D model
2026-06-05

Fixed — Added — Note on the GPU — Verified

The pre-render loop now fans out across CPU cores (multiprocessing.Pool, auto-sized to cpu_count-1, capped Measured on this machine (16 cores): 120 parts 15.4 s serial → 5.26 s on 12 workers = 2.9× faster

0.96.0 THE RESTRUCTURE: monolith → thin shell + `engine/features/` package (+ server hardening)
2026-06-10

Changed — Added — Added — Added

engine/viewer_app.py 2,407 lines → ~330-line shell (config, per-thread SQLite plumbing, RPS init. Same dependency-injection pattern as the earlier extractions (<module>.core = viewer_app) — no import

0.97.0 Search quality + UI dedup finished + the header-wrap layout fix
2026-06-10

Added — Changed

Exact-match boost (C18), did-you-mean (C20), "phrase"/NEAR operators (C22), 60s result LRU (C23); new test_search_quality.py in VERIFY-ALL. UI dedup finished (A2/A3): 12 pages adopt /base.css + /shared.js; 11 inline esc() copies + 12 inline :root palettes stripped. E39 layout fix: the 16-button

0.98.0 Nav consolidation: libraries live in Collections; tools live in a Tools menu
2026-06-10

Changed

Collections is the gateway: LIBRARIES cards for Schematics & wiring + the 3D Library at the top of collections.html; the two standalone header buttons removed (routes untouched, R1). ONE Tools menu replaces nine top-level buttons: Solve it, Part dossier, How to do it, Look-Alike Parts, Circuit Lab +

Rules: R1 backwards-compatible · R2 diagram · R3 dark+PDF · R4 changelog · R5 visual changelog · R6 append-only · R7 legacy track